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Session : Technical Paper Presentation

VIRTUAL GREEN FLEET ORCHESTRATION

Presented By

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INTRODUCTION

Ensuring utilities and fleets unlock the full value of electrification through intelligent, grid-aligned fleet management

2.08Mn+
EVs sold in India in 2024

30%
India's EV sales target by 2030

64%
India's Freight movement is handled by road fleets

- Rapid electrification of commercial fleets is reshaping urban mobility and logistics
- Uncoordinated fleet charging is emerging as a new stress point for local distribution grids, increasing peak demand and operating costs
- EV fleet possess significant distributed energy storage capacity that remains largely untapped
- Virtual Green Fleet Orchestration enables fleets to charge intelligently while supporting grid stability and operational readiness

CONTEXT

India's Accelerating EV Shift

- India is rapidly ramping up EV adoption across logistics, last-mile delivery, and commercial fleets
- This surge is reshaping both mobility patterns and electricity consumption profiles

Peak-hour Fleet Charging Behaviour

- Commercial EV fleets typically return to depots and charge between **6-9 PM,** the system's highest stress window
- This clustered charging increases load spikes, transformer stress, and costs for DISCOMs

Local Distribution Network Impact

- Even small fleet hubs can overload distribution transformer and feeders
- Fast charging installations amplify the strain, requiring grid updates if charging remains unmanaged

Policy & Digital Enablers Unlock Coordination

- Digital platforms enable real-time coordination between fleets, chargers and DISCOMs – making managed fleet charging feasible

RELEVANCE

UTILITIES

India's public charging electricity consumption **more than doubled** from **204 GWh (2022–23)** to **465 GWh (Apr–Oct 2024)**, reflecting rapid load growth utilities must absorb

Peak load demand nationally is projected to rise to **366.4 GW by 2032**, increasing the need for flexible loads like fleets

FLEET OPERATORS

India's EV fleet grew to **6.1 million vehicles by FY2025**, creating strong cost-optimization pressure for operators as energy becomes a major OPEX item

E-commerce and delivery platforms increasingly demand **high vehicle uptime and on-time dispatch**, pushing fleets to adopt **predictive, scheduled charging** instead of manual or ad-hoc practices.

REGULATORS

India had **29,277 public EV charging stations** operational by August 2025—a **6× increase since 2022**, requiring governance of grid-integrated charging behavior

EV charging demand is forecast to reach **38 TWh by 2032**, requiring regulatory frameworks for demand flexibility and VPP participation

WHY NOW?

- By 2030, India's EV growth will be driven predominantly by commercial segments—**delivery fleets, shared mobility, buses, and logistics vehicles**—where unmanaged charging directly impacts operating costs, asset utilization, and grid reliability, making fleet-level orchestration a commercial and systemic necessity
- With projected EV charging demand rising to **3% of total electricity consumption by 2032**, orchestrated fleet charging becomes critical to avoid peak stress and ensure grid reliability.

PRESENTATION ON THE TOPIC (1/2)

What makes VGF Orchestration Transformation?

Convert Fleets into Flexible Energy Assets: EV fleets become dispatchable, flexible energy resources that can modulate load, shift demand, and participate in grid-support programs.

Automated Cost Optimization Across Multiple Layers: Optimizes charging schedules to minimize energy costs using tariff intelligence, peak-shaving, and strategic charge-rate selection.

Human-in-the-Loop Assurance: Automated recommendations are always validated by fleet managers, ensuring transparency, operational trust, and safe decision-making.

Scales Seamlessly Across Fleets and Multiple Depots: Provides a unified orchestration layer that manages diverse chargers, depots, and tariff environments—removing fragmentation in operations.

Designed for India's Operating Conditions: Built to handle local grid constraints, variable tariffs, mixed hardware, and aligns with India's emerging digital energy and EV-integration frameworks.

PRESENTATION ON THE TOPIC (2/2)

Reimagining EV Fleets as Grid Assets

India-Scale, Cloud-Native Fleet & Grid Coordination Platform

VGF functions as a national-scale orchestration layer, connecting EV fleets, chargers, and DISCOMs into a single cloud intelligence fabric—shifting the model from isolated vehicle charging to **system-level coordination.**

Real-Time, Grid-Aware Charging Intelligence

VGF continuously **ingests vehicle SoC, depot load, charger availability, time-of-day tariffs, and local grid congestion signals**—dynamically adjusting charging plans to align fleet operations with grid conditions

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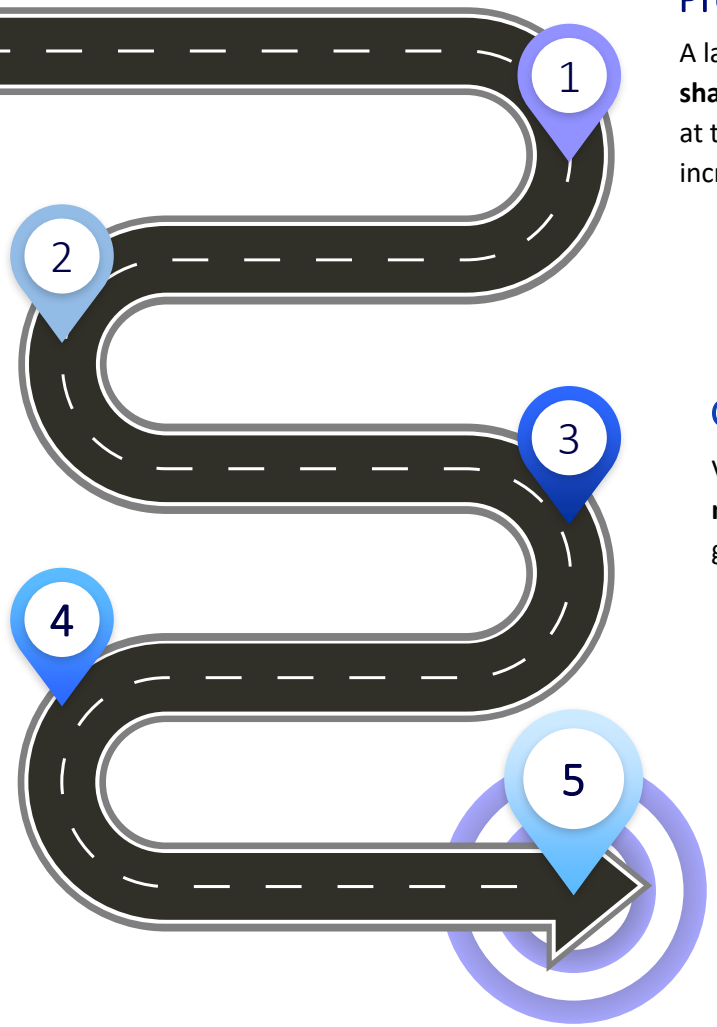
From Unmanaged Load to Coordinated Flexibility

Instead of peak-hour clustering (6–9 PM), VGF automatically staggers and shifts charging across depots—reducing transformer stress, smoothing feeder loads, and avoiding costly grid upgrades.

Fleet-Wide Optimization, Not Vehicle-Level Decisions

Charging is optimized across the entire fleet and depot ecosystem, prioritizing routes, shift schedules, and critical vehicles—while minimizing energy cost and internal congestion.

USE CASE



Problem at a Busy Indian Depo

A large e-commerce logistics depot in Delhi experiences **a sharp evening load spike** when 120+ EV delivery vans plug in at the same time, stressing the local transformer and increasing energy cost.

VGF Onboards the Fleet & Chargers

- Fleet telematics + charger data is connected to VGF (no hardware change).
- VGF begins **tracking SoC, charger availability**, depot congestion, and daily route needs

Grid-Aware Smart Charging

VGF monitors real-time grid demand and tariffs, shifting **non-urgent charging away from peak hours** while guaranteeing morning route readiness.

Fleet Becomes a DER (Flexible Grid Asset)

During high-demand events, VGF **reduces charging load** instantly and can even enable controlled discharge (where allowed), letting fleets support grid stability as a **virtual distributed energy resource (DER)**.

Outcomes for Both Utility & Fleet

Utility: Lower peak stress, deferred infra upgrades, improved reliability
Fleet: Lower energy cost, predictable readiness, and **future revenue potential** through DR/VPP participation

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